Attention head 3 for the agent is attending to a few landmark grid cells, again, probably to establish a sense of position and orientation. Attention head 3 for the key is attending to the door and the door is reciprocally attending to the key.

Putting it all together, we get that the agent is related to the key which is related to the door. If the goal square was in view, we might see that the door is also related to the goal. While this is a simple environment, it has relational structure that we can learn with a relational neural network, and we can inspect the relations that it learns. It is interesting how sparsely the attention is assigned. Each node prefers to attend strongly to a single other node, with sometimes a couple other nodes that it weakly attends to.

Since this is a Gridworld, it is easy to partition the state into discrete objects, but in many cases such as the Atari games, the state is a big RGB pixel array, and the objects we would want to focus on are collections of these pixels. In this case it becomes difficult to map the attention weights back to specific objects in the video frame, but we can still see which parts of the image the relational module as a whole is using to make its decisions. We tested a similar architecture on the Atari game Alien (we just used 4×4 kernel convolutions instead of 1×1 and added some maxpooling layers), and we can see (in figure 10.25) that it indeed learns to focus on salient objects in the video frame (code not shown).

Figure 10.25. Left: Preprocessed state given to the DQN. Middle: Raw video frame. Right: Attention map over state. We can see that the attention map is focused on the alien in the bottom center of the screen, the player in the center, and the bonus at the top, which are the most salient objects in the game.